Department of CSE(AIML), PES University
UE26CS151A -Python for Computational Problem Solving
Laboratory Week 5

Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **3 marks**.

Problem Statements

Q1. Find Missing Numbers

Accept 8 integers from the user, where the numbers are expected to be between 1 and 10. Use sets to find and display the numbers from 1 to 10 that are missing.

Q2. Traverse a Tuple

Write a program to display all elements of a tuple using a loop.

Q3. Convert a List to a Set

Write a program to convert a list into a set

Q4. Find the Difference Between Two Sets

Write a program to create two sets of integers and display the elements that are present in the first set but not in the second set.

Q5. Event Registration Analysis using sets

Write a Python program to accept the names of 8 participants for two different events and store each event's participants in a tuple.

Convert the tuples into sets and display:

1. Participants who registered for both events.
2. Participants who registered only for the first event.
3. Participants who registered only for the second event.
4. The total number of unique participants across both events.

Q6. Set Comparison

Write a program to create two sets of integers and check whether the first set is a subset of the second set. Display an appropriate message based on the result.

Q7. Restricted Access System using sets

Create a set containing 6 authorized access codes. Accept an access code from the user and check whether the code exists in the set. Display "Access Granted" if present; otherwise display "Access Denied".

Q8. Number Frequency and Unique Values

Write a Python program to accept 10 integers from the user and store them in a tuple.

Using a set, identify the unique numbers and display:

1. The original tuple.
2. The set of unique numbers.
3. The numbers that occur more than once.
4. The total number of unique numbers.

Constraint: Do not use the count() method.

Prepared By,
Prof Divya Keerthi